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# Case Study 4: Healthcare & Life Sciences

## OmniLabel AI
## Enterprise Medical Device Label Change Intelligence Platform

Eliminating Silent Scope Omission via Uncertainty-Aware Dual Discovery & 21 CFR Part 11 Audit Ledger

Team Leader: Ashwin Kumar (Lead AI & Systems Engineer)

Competition: Tata Elxsi TELEPORT Season 4

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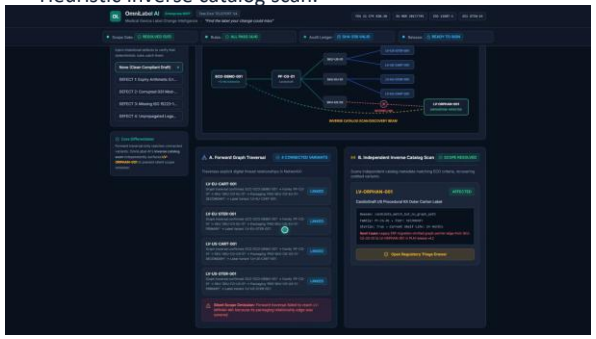
# OmniLabel AI: Enterprise Medical Device Label Change Intelligence

## The Crisis: Silent Scope Omission

- Engineering Change Orders (ECO): Extending sterile shelf life (24m → 36m) requires updating all global packaging.
- Broken PLM/ERP Links: Severed digital threads silently omit unlinked secondary & regional labels.
- Catastrophic Risk: Obsolete labels printed → FDA Class II recalls, 483 warnings, hospital quarantine.

## End-to-End 5-Stage Clinical Pipeline

- 1. Intake & Provenance: Bounded character spans & SHA-256 raw text hash.
- 2. Dual Discovery: Forward thread traversal + Heuristic inverse catalog scan.



## Deterministic Math & Algorithms

- **GS1 GTIN-14 Mod-10 Math:** $S = \Sigma(D_i \times w_i)$, $D_{14} = (10-(S \bmod 10)) \bmod 10$. Rejects invalid barcodes instantly.
- **Inverse Heuristic Scoring:** Score(k,F) = 0.40(Fam) + 0.25(Sterile) + 0.20(Life) + 0.15(Jaccard). Flags cutoff ≥ 0.65.
- **SHA-256 Recursive Ledger:** $H_n = SHA\text{-}256(H_{n-1} \,||\, Event_n \,||\, Payload_n)$. Cryptographically tamper-evident.
- **OpenCV Computer Vision:** Sub-pixel image registration & contour heatmaps.

## 43 / 43 Automated Pytest Suite

✓ 100% Pass Rate: 43 unit, integration & failure-mode stress tests in 3.44s.

✓ Negative Stress Testing: Rejects corrupted GTINs & blocks scope gate bypass.

✓ Tamper Detection: Traps direct SQLite modifications with exact block #.

✓ GitHub CI/CD: Automated multi-stage pipeline (✓ 3/3 green checkmarks).

## The Tripartite Operating Principle

- **Probabilistic AI Extracts:** Bounded spans + SHA-256 raw text provenance.
- **Deterministic Rules Decide:** GS1 Mod-10 checksum math + ISO 15223-1 symbols + Jinja2 SVG (Zero Hallucination).
- **Qualified Humans Sign:** Safe Scope Gate blocks releases until all candidates triaged + 21 CFR 11 e-signatures.

## Regulatory Compliance Matrix

- **FDA 21 CFR Part 11:** §11.10 closed system controls + §11.50 intent-bearing electronic signatures.
- **FDA 21 CFR Part 820:** §820.30/120 labeling controls; eliminates silent scope omission.
- **EU MDR 2017/745:** Annex I safety requirements, expiry arithmetic & GS1 UDI traceability.
- **ISO 15223-1:2021:** Symbol 5.2.3 (Sterile), 5.4.2 (Single-use), 5.1.4 (Expiry) vector geometry checks.

## Business ROI & Integration

- **100% Risk Elimination:** Zero unreviewed labels reach hospitals; avoids Class II recalls.
- **85% Time Savings:** Shrinks ECO packaging review from weeks to minutes.
- **Enterprise PLM Ready:** FastAPI microservice connects to Siemens Teamcenter, Windchill, & SAP.
- **Production Packaging:** Dockerized + React SaaS + 1080p demo walkthrough.

**Team Leader: Ashwin Kumar** | **Project: OmniLabel AI** | **GitHub: https://github.com/Ashwinkumar-k10/omnilabel-ai** | **Release: v1.0.0** | **All 43 Tests Passing**